TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	92.0 ± 0.89
Correlation Stability	94.4 ± 0.66
Distribution Consistency	94.2 ± 0.98
Entropy Preservation	86.5 ± 3.56
Mean Z Score Drift	0.0 ± 0.01
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	30.2 ± 2.40
Data Density	100.0 ± 0.00
Coverage Score	53.8 ± 1.99
Feature Accuracy	84.7 ± 4.47
Novelty Score	0.7 ± 0.02

TabuLLM: AI Benchmark Report

Dataset Audit: thoracic_surgery.csv

1. Dataset Portrait

Attribute	Value
Rows	470
Columns	17
Numeric features	17
Categorical features	0
Missing cells (%)	0.0%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
DGN	3.096	3.152	+0.056	1.8%	OK
PRE4	3.282	3.260	-0.021	0.6%	OK
PRE5	4.569	4.029	-0.539	11.8%	Warn
PRE6	0.781	0.795	+0.014	1.8%	OK
PRE7	0.066	0.074	+0.009	12.9%	Warn
PRE8	0.145	0.133	-0.012	8.1%	OK
PRE9	0.066	0.056	-0.010	15.3%	Warn
PRE10	0.687	0.694	+0.007	1.0%	OK
PRE11	0.166	0.170	+0.004	2.6%	OK
PRE14	11.736	11.782	+0.046	0.4%	OK
PRE17	0.074	0.080	+0.005	7.1%	OK
PRE19	0.004	0.000	-0.004	100.0%	Drift

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	92.0 ± 0.89
Correlation Stability (*2)	94.4 ± 0.66
Distribution Consistency (*3)	94.2 ± 0.98
Entropy Preservation (*4)	86.5 ± 3.56
Mean Z Score Drift (*5)	0.0 ± 0.01
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	30.2 ± 2.40

Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	53.8 ± 1.99
Feature Accuracy (*10)	84.7 ± 4.47
Novelty Score (*11)	0.7 ± 0.02

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8553191489361703	0.8553191489361703	0.8553191489361703	0	
F1	N/A	0.0	N/A	0	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.851063829787234	0.8425531914893616	0.8510638297872342	-1.1102230246251566e-17	
F1	0.037675637056442016	0.08164786967418546	0.04693223443223443	0.009256597375792422	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8531914893617023	0.85	0.8478723404255325	-0.005319148936170215	
F1	N/A	0.03557422969187675	0.016666666666666666	0.016666666666666666	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8553191489361703	0.8521276595744685	0.8521276595744685	-0.003191489361702138	
F1	N/A	0.0	0.013333333333333332	0.013333333333333332	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.848936170212766	0.848936170212766	0.8457446808510639	-0.003191489361702166	
F1	N/A	0.04229102167182662	0.022222222222222222	0.022222222222222222	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.

ACCURACY	0.8138297872340425	0.7893617021276595	0.7851063829787234	-0.028723404255319173	
F1	0.13176356667068742	0.17101603302414666	0.12540960813299643	-0.006353958537690973	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8382978723404257	0.8276595744680855	0.8340425531914895	-0.004255319148936165	
F1	0.07510683760683776	0.08976760082023244	0.11108277343571463	0.035975935828877	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.2106382978723404	0.3531914893617022	0.2297872340425531	0.019148936170212766	
F1	0.256855620661189	0.2059116194443030	0.2553711117710057	-0.001484508890183228	

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.847872340425532	0.851063829787234	0.8510638297872342	0.003191489361702138	
F1	N/A	0.0442647058823529	0.0117647058823529	0.011764705882352941	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8138297872340425	0.8042553191489361	0.8085106382978724	-0.005319148936170215	
F1	0.11972345964178996	0.1090331890331890	0.1285110156915430	0.008787556049753125	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	2.0	2.88	2.16	1.0	0.0	0.0	0.0	1.0	1.0	14.0
2	3.0	3.4	1.88	0.0	0.0	0.0	0.0	0.0	0.0	12.0
3	3.0	2.76	2.08	1.0	0.0	0.0	0.0	1.0	0.0	11.0
4	3.0	3.68	3.04	0.0	0.0	0.0	0.0	0.0	0.0	11.0
5	3.0	2.44	0.96	2.0	0.0	1.0	0.0	1.0	1.0	11.0
6	3.0	2.48	1.88	1.0	0.0	0.0	0.0	1.0	0.0	11.0
7	3.0	4.36	3.28	1.0	0.0	0.0	0.0	1.0	0.0	12.0
8	2.0	3.19	2.5	1.0	0.0	0.0	0.0	1.0	0.0	11.0
9	3.0	3.16	2.64	2.0	0.0	0.0	0.0	1.0	1.0	11.0
10	3.0	2.32	2.16	1.0	0.0	0.0	0.0	1.0	0.0	11.0
11	3.0	2.56	2.32	0.0	0.0	1.0	0.0	1.0	0.0	12.0
12	3.0	4.28	4.44	1.0	0.0	0.0	0.0	0.0	0.0	12.0
13	3.0	3.0	2.36	1.0	0.0	0.0	0.0	1.0	1.0	11.0
14	2.0	3.98	3.06	2.0	0.0	0.0	0.0	1.0	1.0	14.0
15	3.0	1.96	1.4	1.0	0.0	0.0	0.0	1.0	0.0	11.0
16	3.0	4.68	4.16	1.0	0.0	0.0	0.0	1.0	0.0	12.0
17	2.0	2.21	1.88	0.0	0.0	1.0	0.0	0.0	0.0	12.0
18	2.0	2.96	1.67	0.0	0.0	0.0	0.0	0.0	0.0	12.0
19	3.0	2.6	1.68	1.0	0.0	0.0	0.0	1.0	0.0	12.0
20	3.0	2.88	2.48	0.0	0.0	0.0	0.0	0.0	0.0	11.0

B.1 Audit Trace: Fold 1 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	4.04	2.32	2.0	0.0	1.0	0.0	1.0	1.0	12.0
2	3.0	2.96	2.24	0.0	0.0	0.0	0.0	0.0	0.0	11.0
3	3.0	2.96	2.24	1.0	0.0	0.0	0.0	1.0	0.0	12.0
4	3.0	2.0	1.36	0.0	0.0	0.0	0.0	0.0	0.0	11.0
5	3.0	2.36	1.6	1.0	0.0	0.0	0.0	1.0	1.0	11.0
6	3.0	2.72	2.36	0.0	0.0	0.0	0.0	0.0	0.0	11.0
7	3.0	2.68	2.16	0.0	0.0	0.0	0.0	0.0	0.0	12.0
8	4.0	2.32	1.76	1.0	0.0	1.0	0.0	1.0	1.0	11.0
9	2.0	2.72	2.08	0.0	0.0	0.0	0.0	0.0	0.0	12.0
10	3.0	2.32	1.76	0.0	0.0	0.0	0.0	0.0	0.0	12.0

B.2 Audit Trace: Fold 2 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	4.0	2.81	1.88	0.0	0.0	0.0	0.0	0.0	0.0	12.0
2	3.0	2.46	1.76	0.0	0.0	0.0	0.0	0.0	0.0	11.0
3	3.0	5.0	4.96	1.0	0.0	0.0	0.0	0.0	0.0	11.0
4	3.0	2.0	1.8	0.0	0.0	0.0	0.0	0.0	0.0	12.0
5	4.0	2.81	2.24	1.0	1.0	0.0	0.0	0.0	0.0	12.0
6	3.0	2.16	1.8	0.0	0.0	0.0	0.0	0.0	0.0	11.0
7	3.0	2.84	2.24	1.0	1.0	1.0	0.0	0.0	0.0	12.0
8	3.0	3.36	2.8	1.0	0.0	0.0	0.0	1.0	1.0	12.0
9	3.0	2.44	1.96	1.0	0.0	1.0	1.0	1.0	0.0	13.0
10	3.0	5.16	4.28	0.0	0.0	0.0	0.0	0.0	0.0	12.0

B.3 Audit Trace: Fold 3 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	3.4	2.08	0.0	0.0	0.0	0.0	1.0	0.0	12.0
2	3.0	3.68	3.04	0.0	0.0	0.0	0.0	1.0	0.0	12.0
3	4.0	3.44	2.64	1.0	0.0	0.0	0.0	1.0	1.0	12.0
4	4.0	5.12	4.28	0.0	0.0	0.0	0.0	0.0	0.0	12.0
5	5.0	2.38	1.52	1.0	0.0	1.0	1.0	1.0	0.0	12.0
6	2.0	2.72	8.56	1.0	1.0	1.0	0.0	1.0	0.0	12.0
7	4.0	4.24	3.68	1.0	0.0	0.0	0.0	1.0	0.0	12.0
8	3.0	3.87	2.68	0.0	0.0	0.0	0.0	0.0	0.0	12.0
9	5.0	3.67	76.8	0.0	1.0	1.0	0.0	1.0	0.0	12.0
10	3.0	3.8	2.84	1.0	0.0	0.0	0.0	1.0	0.0	12.0

B.4 Audit Trace: Fold 4 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	2.2	1.88	0.0	0.0	1.0	0.0	0.0	0.0	12.0
2	3.0	2.52	2.08	1.0	0.0	0.0	0.0	1.0	0.0	12.0
3	3.0	4.28	3.28	0.0	0.0	0.0	0.0	1.0	0.0	12.0
4	3.0	2.84	66.4	0.0	1.0	1.0	0.0	1.0	0.0	12.0
5	3.0	4.65	4.16	1.0	0.0	0.0	0.0	1.0	0.0	12.0
6	2.0	2.88	2.6	1.0	0.0	0.0	0.0	1.0	0.0	14.0
7	3.0	1.84	1.28	1.0	0.0	0.0	0.0	1.0	1.0	11.0
8	5.0	4.95	4.12	1.0	0.0	0.0	0.0	0.0	1.0	11.0
9	3.0	2.92	2.28	1.0	0.0	0.0	0.0	1.0	0.0	11.0
10	3.0	2.32	1.76	0.0	0.0	0.0	0.0	0.0	0.0	12.0

B.5 Audit Trace: Fold 5 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	2.36	2.08	1.0	0.0	0.0	0.0	1.0	0.0	12.0
2	4.0	2.68	2.0	1.0	0.0	0.0	0.0	1.0	0.0	12.0
3	3.0	3.24	2.52	0.0	0.0	0.0	0.0	0.0	0.0	11.0
4	3.0	2.52	1.72	2.0	0.0	0.0	1.0	1.0	1.0	12.0
5	3.0	4.8	3.48	0.0	0.0	0.0	1.0	0.0	0.0	12.0
6	3.0	2.8	2.12	1.0	0.0	0.0	1.0	1.0	0.0	12.0
7	3.0	3.0	2.44	1.0	0.0	0.0	0.0	1.0	1.0	12.0
8	3.0	2.4	1.88	1.0	0.0	0.0	0.0	1.0	0.0	11.0
9	3.0	4.36	3.92	1.0	0.0	0.0	0.0	0.0	0.0	11.0
10	3.0	3.4	2.8	1.0	1.0	0.0	0.0	1.0	1.0	11.0

B.6 Audit Trace: Fold 6 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	5.0	2.68	2.11	0.0	0.0	0.0	0.0	1.0	0.0	12.0
2	4.0	2.84	2.32	1.0	0.0	0.0	0.0	1.0	0.0	11.0
3	3.0	3.08	2.16	1.0	0.0	1.0	0.0	1.0	0.0	12.0
4	3.0	3.2	2.72	1.0	0.0	0.0	0.0	1.0	0.0	11.0
5	2.0	3.84	2.94	0.0	0.0	0.0	0.0	0.0	0.0	11.0
6	2.0	3.16	2.64	2.0	0.0	1.0	0.0	1.0	1.0	12.0
7	2.0	3.24	2.52	1.0	0.0	0.0	0.0	1.0	0.0	12.0
8	3.0	2.4	1.64	0.0	0.0	0.0	0.0	0.0	0.0	11.0
9	3.0	4.28	4.44	1.0	0.0	0.0	0.0	0.0	0.0	12.0
10	3.0	3.24	52.3	0.0	0.0	0.0	0.0	0.0	0.0	12.0

B.7 Audit Trace: Fold 7 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	2.92	2.16	1.0	0.0	0.0	0.0	1.0	0.0	12.0
2	3.0	4.6	3.28	2.0	0.0	0.0	1.0	1.0	1.0	11.0
3	3.0	3.16	2.6	1.0	0.0	0.0	0.0	1.0	1.0	13.0
4	3.0	3.44	2.92	0.0	0.0	0.0	0.0	0.0	0.0	12.0
5	3.0	1.44	1.36	1.0	0.0	0.0	0.0	1.0	0.0	12.0
6	2.0	2.72	2.08	0.0	1.0	1.0	0.0	0.0	0.0	12.0
7	3.0	2.84	2.16	1.0	0.0	0.0	0.0	1.0	0.0	12.0
8	3.0	2.88	2.48	0.0	0.0	0.0	0.0	0.0	0.0	11.0
9	3.0	2.2	1.44	1.0	0.0	0.0	0.0	1.0	0.0	12.0
10	4.0	5.16	4.96	1.0	0.0	0.0	0.0	0.0	0.0	11.0

B.8 Audit Trace: Fold 8 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	3.6	2.6	1.0	0.0	0.0	0.0	1.0	0.0	12.0
2	4.0	2.72	2.15	0.0	0.0	0.0	0.0	0.0	0.0	11.0
3	3.0	2.92	73.3	0.0	1.0	0.0	0.0	0.0	0.0	12.0
4	3.0	3.44	3.13	1.0	0.0	0.0	0.0	1.0	0.0	12.0
5	3.0	3.12	2.73	2.0	0.0	0.0	0.0	1.0	1.0	11.0
6	3.0	3.0	2.64	1.0	1.0	0.0	0.0	1.0	0.0	11.0
7	3.0	2.76	2.28	0.0	0.0	0.0	0.0	0.0	0.0	11.0
8	4.0	5.03	79.3	1.0	0.0	0.0	1.0	0.0	0.0	11.0
9	2.0	2.16	1.56	1.0	0.0	0.0	0.0	1.0	0.0	11.0
10	3.0	2.0	1.0	1.0	0.0	1.0	0.0	1.0	1.0	11.0

B.9 Audit Trace: Fold 9 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	2.83	1.88	1.0	0.0	0.0	0.0	1.0	0.0	12.0
2	3.0	3.88	3.08	1.0	0.0	0.0	0.0	1.0	0.0	11.0
3	3.0	2.52	2.08	1.0	0.0	0.0	0.0	1.0	0.0	12.0
4	3.0	4.0	2.56	1.0	0.0	0.0	0.0	1.0	0.0	11.0
5	3.0	4.56	2.68	1.0	0.0	0.0	0.0	1.0	0.0	11.0
6	4.0	3.28	3.68	0.0	0.0	0.0	0.0	0.0	0.0	12.0
7	3.0	3.44	3.13	1.0	0.0	0.0	0.0	1.0	1.0	12.0
8	3.0	3.64	3.12	1.0	0.0	0.0	0.0	1.0	0.0	12.0
9	3.0	2.56	60.9	0.0	0.0	0.0	0.0	0.0	0.0	11.0
10	3.0	2.8	3.36	1.0	0.0	0.0	0.0	1.0	0.0	12.0

B.10 Audit Trace: Fold 10 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	2.84	2.36	1.0	1.0	1.0	0.0	0.0	0.0	12.0
2	3.0	2.48	1.6	1.0	0.0	0.0	0.0	1.0	0.0	12.0
3	3.0	3.24	2.4	1.0	0.0	0.0	0.0	1.0	0.0	11.0
4	3.0	4.2	3.32	0.0	0.0	0.0	0.0	0.0	0.0	12.0
5	3.0	5.16	4.28	1.0	0.0	0.0	0.0	0.0	0.0	12.0
6	5.0	3.52	2.08	0.0	0.0	0.0	0.0	0.0	0.0	12.0
7	3.0	3.48	2.52	1.0	0.0	0.0	0.0	1.0	0.0	14.0
8	3.0	4.48	3.48	0.0	0.0	0.0	0.0	0.0	0.0	12.0
9	3.0	2.63	67.3	1.0	0.0	0.0	1.0	1.0	0.0	11.0
10	4.0	5.16	4.96	1.0	0.0	0.0	0.0	0.0	0.0	11.0